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Ancient DNA analysis of dental calculus

Laura S. Weyrich [a], Keith Dobney [b], Alan Cooper [a, *]

[a] The Australian Centre for Ancient DNA, The University of Adelaide, Adelaide, Australia
[b] Department of Archaeology, School of Geosciences, University of Aberdeen, Aberdeen, UK

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ABSTRACT

Dental calculus (calcified tartar or plaque) is today widespread on modern human teeth around the world. A combination of soft starchy foods, changing acidity of the oral environment, genetic predisposition, and the absence of dental hygiene all lead to the build-up of microorganisms and food debris on the tooth crown, which eventually calcifies through a complex process of mineralisation. Millions of oral microbes are trapped and preserved within this mineralised matrix, including pathogens associated with the oral cavity and airways, masticated food debris, and other types of extraneous particles that enter the mouth. As a result, archaeologists and anthropologists are increasingly using ancient human dental calculus to explore broad aspects of past human diet and health. Most recently, high-throughput DNA sequencing of ancient dental calculus has provided valuable insights into the evolution of the oral microbiome and shed new light on the impacts of some of the major biocultural transitions on human health throughout history and prehistory. Here, we provide a brief historical overview of archaeological dental calculus research, and discuss the current approaches to ancient DNA sampling and sequencing. Novel applications of ancient DNA from dental calculus are discussed, highlighting the considerable scope of this new research field for evolutionary biology and modern medicine.
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An introduction to dental calculus

In the absence of modern dentistry and daily brushing, dental plaque, a biofilm of oral bacterial species, builds up on the surface of teeth, calcifying into supra- (above) and sub- (below) gingival calculus (Fig. 1) (White, 1997). As calcium phosphate mineral salts deposit on the tooth surface, the bacterial biofilm becomes preserved, trapping living and non-viable oral microorganisms. The microorganisms within calculus represent species identified in saliva and dental plaque, and comprise over 600 different taxa (Dewhirst et al., 2010; Liu et al., 2012). In addition to oral bacterial species, viral and fungal taxa, remnant food debris, and disease-causing microorganisms isolated from the upper and lower respiratory tract have also been identified in both ancient and modern dental calculus (Henry and Piperno, 2008; Henry et al., 2011; Warinner et al., 2014). As a result, dental calculus represents the first accurate fossilised record of bacterial communities associated with the human body (microbiome), and an important source of information about the evolution and interplay of the human microbiome, diet, and health.

Typically, calculus forms on the tooth surface adjacent to salivary ducts (White, 1997), although the amount and distribution across the dentition can be dependent on poor oral hygiene, genetic pre-disposition, or diet, with the consumption of soft, sticky carbohydrates leading to the accumulation of large deposits (Arensburg, 1996; Lieverse, 1999; Al-Zahrani et al., 2004). As a result, calculus is nearly ubiquitous in modern, post-agricultural populations, while it can be more rare amongst ancient and modern foraging or hunter-gatherer groups, likely due to dietary differences (Aufderheide and Conrado, 1998). It can sometimes be difficult to find calculus on fossil and archaeological material collected in the past, as it was as it was formerly a common practice to remove calculus from ancient teeth to investigate dental wear patterns, complete oral measurements, or collect additional evidence of oral disease. Subsequent drying and shrinkage of the calcified matrix during museum storage can also result in separation and loss of calculus specimens post-mortem.

Historical context of calculus research

During the last few decades, dental calculus has been recognised as an informative tool to understand ancient diet and health. In the 1970's, the first studies of ancient dental calculus took place on archaeological samples of cattle, sheep, and horse teeth (Armitage,

* Corresponding author.
E-mail address: Alan.cooper@adelaide.edu.au (A. Cooper).

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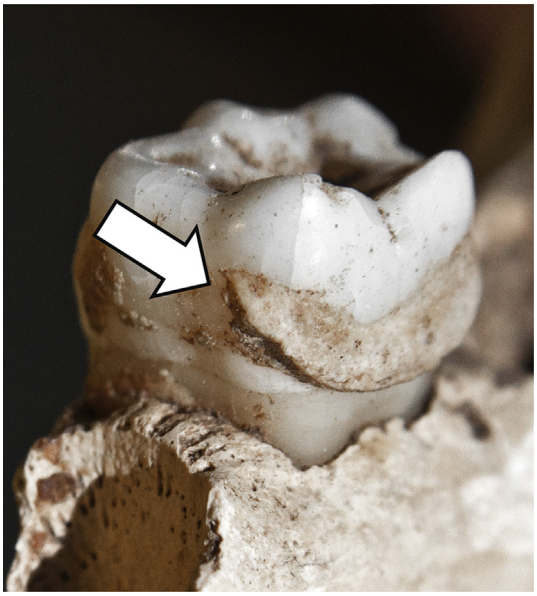


Figure 1. Supra-gingival dental calculus is identifiable in a concave ring on a lower molar from a Medieval specimen, York, UK.

1975) and revealed the presence of numerous opal phytoliths (silica content of some plant cells). These early proof-of-concept studies provided the foundations for future calculus approaches and highlighted the potential for direct dietary analysis from ancient dental calculus, as well as concerns about microscopic contamination of samples with external materials.

During the following decades, archaeological dental calculus from domestic animals was used to identify trapped plant phytoliths, as well as develop strict decontamination protocols that included washing, ultraviolet (UV) irradiation, and bleach treatments. These animal calculus studies recovered the well preserved organic remains of pollen grains, animal hair, and a diverse range of unidentifiable animal and plant tissues (Dobney and Brothwell, 1986, 1988), and developed a systematic protocol for recording and quantifying dental calculus present on archaeological domestic bovid teeth (Dobney and Brothwell, 1987). Scanning electron microscopy (SEM) of archaeological human and animal dental calculus samples was used to further identify food particles and to record the first observation and description of in situ calcified oral microbes (Dobney and Brothwell, 1988). Further studies revealed the extensive and ubiquitous presence of well-preserved calcified microorganisms in human calculus, and formed the first investigations into the ancient oral microbiome (Hansen et al., 1991; Dobney, 1994).

Subsequent work on dental calculus discussed the impacts of ancient diets on dental calculus formation (Lieverse, 1999), and rekindled interests in using food particles trapped in dental calculus to answer anthropological questions targeted at diet, environment, and disease. Over the last decade and a half, starch granules, phytoliths, and pollen obtained from dental calculus have been used as a basis for exploring both the diet and paleoenvironment of ancient humans, Neanderthals, and even *Australopithecus sebida* (Henry and Piperno, 2008; Piperno and Dillehay, 2008; Wesolowski et al., 2010; Henry et al., 2011, 2012; Li et al., 2013). Microfossil analysis has more recently been combined with proteomic and small compound analysis to provide more accurate depictions of entrapped dietary information within calculus (Hardy et al., 2012; Warinner et al., 2014).

In the past 30 years, the use of dental calculus in anthropological and archaeological research has advanced significantly. However, assessing diet through starch grain and phytolith analysis, even using modern tools, remains problematic, and technical methods used in these studies are additionally quite different than those suggested in earlier studies. For example, many starch grains and phytoliths can only be identified to the family or order level due to structural similarities amongst vastly different species (Hardy et al., 2009). Furthermore, phytolith analysis has been performed via submerging calculus fragments in water, without rigorous decontamination or de-calcification of this complex matrix (Henry et al., 2012). Key questions remain about reproducibility and accuracy of results, as well as the difference between innate dental calculus particles and the levels and contributions of contamination from soil through time. While microstructural analysis of dental calculus has proved to be a fruitful research avenue, there is a clear need for the standardisation of techniques and analysis.

Initial ancient DNA analyses of dental calculus microbes

The genetic analysis of ancient dental calculus holds considerable potential for both the study of dietary information and the microorganisms preserved within the calculus itself. Scanning electron microscopy analyses have shown that whole bacteria are preserved within dental calculus (Dobney and Brothwell, 1988; Dobney, 1994; Vandermeersch et al., 1994; Pap et al., 1995; Arensburg, 1996), revealing rods and cocci of several different bacterial phyla. However, the overall taxonomic resolution, diversity, and composition cannot be resolved utilizing morphological techniques alone.

The first demonstration that bacterial DNA remained intact within preserved calculus (Preus et al., 2011) was quickly followed by successful DNA extraction and selected sequencing of ancient oral microbial DNA (De La Fuente et al., 2012). Targeted polymerase chain reaction (PCR) was used to amplify genes present in five different oral bacterial species, demonstrating that dental calculus provides a preserved bacterial fossil record of human associated microbes. In the following year, the first attempts to understand changes in oral microbial community structure and overall species diversity through time applied High Throughput Sequencing (HTS) and metagenomic methodologies (Adler et al., 2013). This study used a large number and variety of samples to provide detailed views of changes in bacterial community diversity and the oral microbiome over an 8000 year time frame. In addition to revealing long-term dietary impacts on health and disease and the presence of specific bacterial pathogens through time, this study also demonstrated that bacteria can provide a distinct cultural fingerprint. Similar to gut microbial communities, oral bacterial communities can adapt according to different diets, hygiene, or environmental practices, generating specific cultural signals. This discovery provided an important new means to track human migration and admixture history, and confirmed the broader importance of analyses of dental calculus for fields such as anthropology and archaeology.

More recently, another metagenomic sequencing strategy was applied to ancient dental calculus (Warinner et al., 2014). Shotgun sequencing was utilised to examine four Medieval specimens from Germany, providing greater resolution of oral microbial communities earlier identified by 16S rRNA sequencing (Warinner et al., 2014). In addition, the study revealed information regarding the presence of antibiotic resistance genes and key virulence genes present in oral microorganisms from Medieval Germany. Decontamination methods, such as EDTA washing and bleach treatment, also did not inhibit the ability to obtain DNA from dental calculus and quantification of DNA abundance, demonstrating that dental calculus can contain up to 1000 times more DNA than bone from the same individual (Warinner et al., 2014), further highlighting the

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massive potential for dental calculus to revolutionise ancient biomedical research.

Current protocols for ancient DNA analysis of dental calculus

Sampling

In general, a calculus deposit remains separate from the enamel surface of the tooth, making it possible to simply pry or apply pressure to dislodge the calculus deposit (Henry and Piperno, 2008). Typically, the largest deposits of dental calculus form on the supra-gingival surface of the molars and premolars on the lingual side of the mandibular teeth and the buccal side of maxillary teeth due to the proximity to salivary ducts. In general, the largest deposit of calculus on a specimen is used as the sample, and this is likely to be supra-gingival calculus on molars. While sub-gingival calculus occurs, it is often associated with gingival recession and has been linked to periodontal disease (reviewed in Timmerman and van der Weijden, 2006). Supra- and sub-gingival calculus bacterial communities can be quite different, and researchers should be aware of this potential bias (Ximénez-Fyvie et al., 2000).

The calculus sample should be collected from the specimen (isolated tooth, skull, or jaw) over clean aluminium foil, to catch fragments of the calculus as it is removed from the teeth. The foil can also be wrapped closely around the sampling area to prevent calculus samples flying away from the sampling area, as considerable lateral force is required to dislodge some samples. The thickest or most pronounced edge of the calculus should be identified and a dental pick or dissecting needle used to apply pressure on the edge, directly adjacent to the tooth enamel, until the calculus breaks free. The use of large metal implements should generally be avoided, due to the risk of scratching or damaging the tooth surface. Often, the calculus deposit will fragment during this process, but the pieces should be caught below on the aluminium foil.

Once the calculus is removed from the tooth, the sample on the aluminium foil should be collected and tipped into a small labelled, sterile plastic bag or screw cap tube. Movement within plastic or glass tubes during transit may break brittle samples, and if tubes must be used, plastic ones are recommended as both the glass tubes and the samples inside can be easily crushed. Each sample bag or tube should be carefully labelled with the larger specimen context, sample information, and date. In addition, detailed notes should be taken about the collection, collector's name, and information about the host specimen, such as which tooth, region, and aspect were sampled, how many teeth of the sampled individual had calculus, abnormalities associated with sampling site or tooth, and any other cultural or environmental information. For example, chewing betel nut can stain dental calculus a dark brown colour, and recording information such as this can be useful during the extraction processes and data interpretation.

Decontamination and DNA contaminant control

The rugose, pitted, and porous nature of dental calculus, along with the risk of soil contamination or bacterial contamination in laboratory reagents, makes dental calculus one of the most challenging ancient DNA samples to examine. Rigorous decontamination procedures must be followed to remove or irreparably damage contaminating DNA on the outer surface of the calculus sample (Cooper and Poinar, 2000). Ideally, one large fragment of calculus ($>2 \times 2$ mm) should be selected for analysis rather than powder or multiple smaller fragments, because the surface of a single large sample can be more easily decontaminated. The remaining fragments can be utilised to replicate results. Decontamination should occur immediately before DNA extraction, as some bacterial and fungal contaminants can grow even at refrigerated temperatures.

Sample decontamination should involve UV irradiation (e.g., >15 min on each surface); the sample can be manipulated during this process with sterile tweezers that are ideally pre-treated with 4% bleach and UV irradiated. Following UV irradiation of the calculus sample, bleach treatment is strongly recommended to decontaminate beyond the exterior surface. The sample can be submerged in a sterile container (petri dish) containing molecular grade bleach (4% final dilution) for five minutes and then rinsed by submersion in a separate sterile container of molecular grade ethanol (90%) to remove traces of bleach, for up to three minutes. Bleach and ethanol reagents should be freshly prepared and diluted using only sterile, molecular grade water. Any water aliquots should be frozen prior to use to prevent microbial growth. The sample should be allowed to air dry for up to five minutes on a sterilised surface or in a sterile container (petri dish), before directly proceeding to pulverisation and extraction.

Pulverisation

There are multiple ways to pulverise ancient material, but a simple method currently used to powder calculus at the Australian Centre for Ancient DNA is to smash the sample with a hammer. The decontaminated sample is placed in the bottom corner of a clean, sterile plastic bag (3 cm $\times$ 4 cm or smaller), and hit lightly and repeatedly with a pulverising hammer. Once the sample is powdered, the opposing corner of the bag is removed with sterile scissors, and the sample is poured into a clean, sterile screw-cap tube for extraction. Creased aluminium foil can be used to assist the transfer into the tube if necessary. Sterile tubes without calculus powder should also be incorporated into the process at this step, as a control to monitor the DNA content present in lab materials and the environment.

Extraction

Currently, several extraction protocols are available for ancient DNA analysis of dental calculus (Kuczynski et al., 2012; Adler et al., 2013; Warinner et al., 2014), while new methodologies are being developed to allow phytolith, protein, and DNA analyses to be performed concurrently from a single calculus sample. Most ancient DNA extraction protocols recommend decalcification as the first step; ethylenediaminetetraacetic acid (EDTA) is utilized to decalcify ancient bone or teeth (Pääbo, 1989; Shapiro and Hofreiter, 2012), although this can limit downstream proteomic and isotopic analysis. Decalcification of dental calculus can be completed by placing the sample in a mixture of EDTA, sodium dodecylsulfate (SDS), and proteinase K and incubating for 24 h at 55 °C (Willerslev and Cooper, 2005; Brotherton et al., 2013). For difficult samples, DNA yields may be improved through longer incubations at lower temperatures (i.e. 48 h at 37 °C), increased concentrations of EDTA, or repeating the process (Orlando et al., 2011).

Once the sample is decalcified, there are several DNA extraction methods available. Commercial DNA extraction kits may be used if the sample is well preserved and not ancient in origin, although the EDTA from the de-calcification steps will need to be removed before the sample is compatible with most DNA extraction kits (Adler et al., 2013). Several kits have been optimized to minimise laboratory contamination and are primarily used in modern DNA studies of the human microbiome, such as the UltraClean Microbial DNA Isolation Kit (MoBio) or DNA Investigator Kit (Qiagen) (Kuczynski et al., 2012). However, poorly preserved or ancient samples generally require more rigorous extraction methods (Hoss and Pääbo, 1993; Dabney et al., 2013). In-house silica-based

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methodologies are common in the ancient DNA field and appear to be several orders more effective at retrieving DNA from dental calculus samples than kits (Hoss and Pääbo, 1993) (Weyrich, Personal experience).

Extraction blank controls must always be run concordantly with ancient samples to monitor and identify laboratory contamination (Cooper and Poinar, 2000; Pääbo et al., 2004; Mühl et al., 2010). Extraction blank controls should be created by processing blank sample tubes alongside calculus samples during the DNA extraction process, allowing researchers to obtain potential contaminant bacterial DNA from reagents and the laboratory environment. A control is typically included as the first and last sample in the procedure, as well as regularly between samples, providing information on both the contamination within the laboratory reagents and plastic-ware, and the potential cross-contamination between samples during the extraction procedures. Sequences identified within extraction blank controls can then be bioinformatically removed from the pool of sequences identified in ancient samples.

DNA sequencing strategies

Sequencing strategies include targeted amplification, amplicon, and shotgun sequencing. Targeted amplification of specific bacterial sequences has been successfully applied to dental calculus with species-specific primer sets used to identify five different oral bacterial species (De La Fuente et al., 2012). While inexpensive, this technique limits the number of DNA sequences that can be identified simultaneously and can be labour intensive if numerous genes or increased DNA fragment length are desired. However, this method can be useful to identify single species, verify the presence of certain organisms, test the DNA preservation of the sample, or quickly detect some specific genes of interest.

Amplicon sequencing or metabarcoding has also successfully been applied to dental calculus samples (Adler et al., 2013). This method amplifies a gene region that is shared or conserved amongst numerous target species and can provide taxonomic resolution, therefore identifying hundreds of species within a single PCR amplification. For example, amplicon libraries that targeted the fourth variable (V4) region of the 16S ribosomal RNA gene in bacteria identified over 800 different oral species in a single calculus sample (Adler et al., 2013). This strategy is particularly important when investigating microbial community structure or when the specific expected species are unknown (Chakravorty et al., 2007). This strategy can provide sequences for thousands of taxa present within a sample, although species level resolution may not be obtained in all cases (Jumpstart Consortium Human Microbiome Project Data Generation Working Group, 2012). Compared with targeted amplification, this technique is more expensive, as it requires HTS sequencing runs. A significant knowledge of bioinformatics and metagenomic sequence processing is also required to efficiently interpret results.

Amplicon libraries can be designed and amplified for any target gene of interest, such as fungal (*its*), eukaryote (18S ribosomal RNA), and dietary components, e.g., plant (*trnL*) or animal (*rbc* or 16S mammalian ribosomal RNA), present within the calculus (Taberlet et al., 2007; Dunshea, 2009; Murray et al., 2011; Parfrey et al., 2011). In addition to bacteria, additional microbial groups, such as protists and fungi, are likely to reveal further changes associated with large dietary and cultural shifts throughout history. However, the most powerful synergy will be when the metabarcoding of plant and animal markers can be applied efficiently to dental calculus alongside bacterial and fungal markers, revealing how microbial communities and human health have changed in response to specific dietary inputs.

Shotgun sequencing of dental calculus can be used to obtain a random sampling of DNA fragments present, i.e., bacteria, fungi, viruses, plants, and animals. Unlike the previous strategies that target a single genomic region for amplification prior to sequencing, this procedure attaches adapter sequences to any available double (Meyer and Kircher, 2010) or single stranded DNA molecule (Meyer et al., 2012). The adapter sequences can then be used to amplify all of the DNA molecules in the resulting genomic library, regardless of the species or taxonomic resolution of the fragment. The adaptors also generally contain recognition sites for particular sequencing machines, allowing the libraries to be rapidly sequenced. Shotgun sequencing has the ability to detect microbial, viral, and dietary information simultaneously, although at a much lower level of coverage due to the enormous metagenomic space being traversed. Furthermore, downstream data analysis can be more challenging, as many genes are shared amongst species and remain unidentifiable to any specific taxa (Prakash and Taylor, 2012). Typically, shotgun sequencing is also the most expensive, as numerous genes from a single species are required to identify specific taxa. However, this approach also permits the retrieval of sequences without any a priori knowledge, such as viruses or short fragments that could not be amplified with targeted strategies. Shotgun sequencing also provides functional metagenomic information, e.g. increased or decreased presence of antibiotic resistance genes through time. Furthermore, shotgun libraries can also be enriched for target sequences using hybridisation capture with DNA or RNA probes. Such hybridisation capture approaches can be used to characterise draft genomes of ancient microbes, and are likely to be applicable to trapped oral microbes in calculus (Hodges et al., 2009; Burbano et al., 2010).

Future research avenues

Dental calculus has the potential to be one of the most valuable samples obtainable from an ancient skeleton, providing information on the diet, health, diseases, microbes, environment, and perhaps even cultural affinity of an individual. At the moment, high throughput DNA sequencing is the standard approach to investigate bacterial communities living in the mouths of ancient humans. However, several additional important research areas are likely to emerge from ancient DNA analysis.

Dietary analysis

Although microfossil analysis of dental calculus has provided valuable information about the dietary behaviour of different cultures and provided comparisons between hominid groups, difficulties in specific identifications and the interpretation of data has left many questions unanswered (Collins and Copeland, 2011). Ancient DNA analysis has the potential to considerably enhance the resolution of specific plant and animal species locked within dental calculus and identify how this changes through time and between cultural groups or species. Amplicon libraries of *trnL* or 18 rRNA sequences can provide an overview of plant species, which can then be independently verified by targeted amplification of species-specific sequences. DNA analysis should also be able to confirm or refute earlier microfossil and protein analyses of dental calculus, providing species resolution of starch grains or microfossils (Salazar-García et al., 2013; Warinner et al., 2014) or identifying plants that contributed specific protein biomolecules to calculus (Hardy et al., 2012). Alternatively, dental calculus may also serve as an excellent fossil record for plants, preserving early DNA sequences of domesticated (or wild) plants present in ancient teeth, e.g., from early farmers.

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Microbial and disease evolution

Although we have focused on ancient DNA techniques that identify all species present in a sample, larger genomic sequences of single species, including draft bacterial, fungal, or viral genomes, will likely be major areas for future dental calculus research. Draft genomes of bacterial pathogens have already been obtained from bone or teeth through hybridisation enrichment, including the causative organisms for leprosy, tuberculosis, and plague (Bos et al., 2011; Bouwman et al., 2012; Schuenemann et al., 2013). The high density of bacterial DNA in dental calculus suggests that it will be possible to generate genomes of oral and respiratory pathogens, providing insight into the evolution of diseases, mechanisms of pathogen transmission through time, and paleoepidemiology. A key area will be to examine the changes in bacterial community structure and functionality, which may reveal selection pressures and mechanisms of bacterial evolution through time.

Monitoring changes in the human microbiome

It is essential that we understand how our commensal microorganisms (microbiome) have changed through time, as this could provide insights not only for anthropologists and archaeologists, but also for medical researchers, microbiologists, and evolutionary biologists. Numerous diseases, including obesity, diabetes, arthritis, and autism, have been linked to alterations of the human microbiome (reviewed in Cho and Blaser, 2012). As a result, bacterial 16S or eukaryotic 18S rRNA amplicon sequencing of ancient dental calculus can be utilised to monitor changes and alterations in the human microbiome through time, to allow correlation with changes in human immunogenetics, population size and dynamics, dietary inputs, and environmental and social change. Work such as this has already revealed major shifts in the human microbiome during large cultural and environmental changes (Adler et al., 2013; Harper and Armelagos, 2013), although the specific mechanisms underlying these changes require further investigation. If researchers can identify mechanisms and species associated with key biocultural changes in the past, then the same principles might be used to develop medical solutions in the future.

The human microbiome as a microbial fingerprint

Modern DNA studies have been able to link the microbiome of an individual with objects that have been touched, e.g., a computer keyboard (Fierer et al., 2010). Somewhat analogously, different ancient cultures appear to possess unique bacterial signals based on their overall community structure (Adler et al., 2013). These observations suggest that ancient DNA analysis of dental calculus can be used to track humans and populations through space and time, as they enter and leave specific cultural regions. Furthermore, unique bacterial species are likely to be associated with specific dietary inputs (Hehemann et al., 2010), so certain microbes could provide proximal information, suggesting where an individual has travelled or if individuals had different diets, even if the DNA from the dietary input itself is not identified.

Conclusions

While previous analysis of dental calculus has provided a wealth of anthropological information, ancient DNA techniques promise to open a broad new field of research. If environmental and laboratory contamination can be controlled, then targeted amplification, amplicon libraries, and shotgun sequencing approaches will continue to deliver unique genetic evidence about the palaeoanthropological record. The combination of approaches will provide new investigative power to view and understand the microscopic world associated with the human body, providing insight into the evolution, movement and demise of ancient societies, new ways to investigate the mechanisms and impacts of bacterial evolution through time, and to monitor how cultural, environmental and social changes impact human health and behaviour. Although the anthropological and archaeological study of dental calculus has been advancing for the past three decades, the paradigm-shift initiated through ancient DNA research is underway, allowing researchers to address key questions and ask new ones.

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